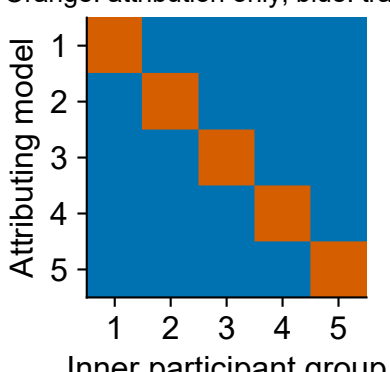
Participant p
Outer test only

Other 42 participants
5 inner grouped folds
Training-only scaling
Train inner XGBoost

Inner-held-out attributions
Normalize electrode scores
Median / MAD ranking

Select top k -> retrain reduced model on outer training participants -> test only on p

Orange: attribution only; blue: training



Electrode magnitude: mean |TreeSHAP|
Fold score: normalize across electrodes
A = median normalized magnitude
U = median absolute deviation
Q(lambda) = A - lambda U

Report every fixed lambda; do not tune on p